



Data Article

A dataset of blood slide images for AI-based diagnosis of malaria



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ABSTRACT

Malaria is a major public health challenge in sub-Saharan Africa. Timely and accurate diagnosis of malaria is vital to reduce the caseload and mortality rates associated with malaria. The use of microscopy in malaria screening is the gold standard recommended method by the World Health Organisation (WHO). In Uganda, utilization of microscopy is challenged by insufficient expertise to interpret the images accurately, affecting the efficiency, effectiveness and accuracy of malaria detection and diagnosis. We present a benchmark dataset of thick and thin blood smear images for automatic malaria screening in Uganda. Mobile Microscopy data was collected from Mulago Hospital, Department of Internal Medicine, Makerere University and Kiruddu National Referral Hospital in Uganda. The labelled image data can be used to build computational models implemented with convolution neural networks. The dataset has 3000 labelled thick blood smear images and 1000 labelled thin blood smear images. The datasets will support robust and accurate deep learning

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models for malaria diagnosis using thick and thin blood smear images with reasonable detection accuracies.

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Specifications Table

Subject	Applied Artificial Intelligence, Computer Vision and Pattern Recognition, Medical Imaging
Specific subject area	Object Detection for malaria microscopy with thick and thin blood slide images
Data format	Raw, Filtered
Type of data	Image, Text
Data collection	The image data was collected using microscopes and smartphones. The microscope used for the data collection was the Olympus CX-23 ¹ optical microscope. The phones used to capture the images were Samsung S8+, Redmi 8 and Redmi 9. The images were taken from randomly selected slides, with a maximum limit of fifty (50) images per slide for the thick blood slides and a maximum limit of ten (10) images per slide for the thin blood slides.
Data source location	The data was collected from Mulago Referral hospital and Kiruddu National Referral Hospital in Kampala, Uganda.
Data accessibility	Repository name: Harvard Dataverse: Lacuna Malaria Datasets Data identification number: 10.7910/DVN/VEADSE Direct URL to data: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VEADSE Instructions for accessing these data: Visit the dataset link and download file objects from the repository. The objects are either RAR files for the images or PDF for the data sheets.
Related research article	J.A. Quinn, A. Andama, I. Munabi, F.N. Kiwanuka. Automated Blood Smear Analysis for Mobile Malaria Diagnosis. Chapter in Mobile Point-of-Care Monitors and Diagnostic Device Design, eds. W. Karlen and K. Iniewski, CRC Press, 2014.

1. Value of the Data

- The dataset includes thick and thin blood slide images captured from a microscope using a smartphone camera. The images are annotated to show different objects, such as malaria parasites and white blood cells, and this dataset can be used to develop automated solutions for mobile microscopy.
- Malaria parasite detection: The dataset can be used to develop machine learning models for detecting malaria parasites from images of thick and thin blood slides, a key malaria microscopy procedure for malaria diagnosis. Our dataset addresses a critical need in malaria microscopy for automating the detection of small parasites in blood smears with state-of-the-art machine learning models, a procedure that is currently a strenuous manual procedure.
- Malaria parasite growth stage detection. Our thick blood smear datasets contain various parasite growth stages/life cycles i.e. gametocytes and trophozoites. The gametocytes indicate the spread stage of malaria while the trophozoites are the malaria parasites. The dataset can, therefore, be used to build machine learning models for automated detection of malaria parasite growth stages.
- Malaria parasitaemia estimation. Our thick and thin blood smear datasets can also be used for malaria parasitaemia determination using state-of-the-art machine learning methods, similar to work in [8]. Manually counting white blood cells and parasites for parasitaemia determination is extremely time-consuming and tedious. Through detecting and quantifying

¹ [https://www.olympus-lifescience.com/en/microscopes/upright/cx23/#!cms\[focus\]=cmsContent1392](https://www.olympus-lifescience.com/en/microscopes/upright/cx23/#!cms[focus]=cmsContent1392).

trophozoites and White blood cells on thick smears, various machine-learning algorithms can be used for malaria parasitaemia estimation with this dataset. Determination of malaria parasitaemia quantifies the level of infection of malaria.

- Classification of the parasitized and uninfected cells. Due to the plasmodium parasite, malaria is transmitted mainly through red blood cells. Machine learning models will be developed to analyze blood samples to determine whether the cells are parasitized or not. Our thin blood smear image dataset will be utilised to support the development of machine learning models.
- Artefact analysis: Local microscopy is reported to have a wide range of poor specificity compared to expert microscopy. Poor blood film preparation generates artefacts commonly mistaken for malaria parasites, including bacteria, fungi, stain precipitation, and dirt and cell debris. Standard blood components such as platelets also confound malaria diagnosis. The thin blood smear dataset can be used to investigate artefact detection based on machine learning models. Therefore, in this study, we expand the dataset not only in the increasing number of image datasets but also to the best of our knowledge, this is the first mobile smartphone based thin blood smear image dataset that is publicly available for Uganda.

2. Background

The shortage of highly skilled microscopists primarily contributes to the burden of malaria diagnosis in Africa [3,7]. Towards improving the gold standard, we have in the past confronted automated microscopy of malaria by exploiting recent technological advances in 3D printing to enable the development of a low-cost 3D printed adapter that enables attachment of a wide range of smartphones on a microscope and deployment of deep learning algorithms for pathogen detection [1]. This has enabled the production of adequate hardware and software for data collection and disease diagnosis, respectively.

Although earlier studies [1,2,4–6] have demonstrated that malaria can ably be diagnosed through AI on thick blood smear images, this research including the data collection procedure has been ad-hoc in its current state and is limited. Certain variations in the datasets at phone and microscope level were not captured and standardised like phone resolutions and grid reading to allow for realistic field context. Thus, the detection results obtained were not reproducible across different experimental settings. This dataset, therefore, provides additional standardised datasets that capture these different features and a new dataset of thin blood smear images from Uganda, which have not been available and have not been widely studied to support improved automated malaria microscopy.

3. Data Description

The dataset contains images and annotations showing different objects of interests. The image dataset is categorized into thick and thin blood slide images which form the main folders in the dataset. Under each of these categories are two subfolders containing images and annotations. The image subfolder contains the raw images of the blood slides as captured by a phone and the labels folder contains text files with annotations for the images (Table 1).

Table 1
Number of images and labels in the dataset.

Folder	Images	Labels
Thick	3000	3000
Thin	1000	1000

4. Experimental Design, Materials and Methods

4.1. Site and sample selection

The data was collected from Mulago National Referral and Kiruddu Hospital, major medical facilities in central Uganda. These sites were selected because they serve a large population from the central region of Uganda and provide diagnosis of several diseases, including malaria.

The blood slides used in this study were previously collected samples using the process described in [10] by the lab technologists and were stored awaiting disposal. We selected slides in the pre-disposal stage to ensure that the samples cannot be traced back to patients, hence guaranteeing patient privacy. Despite being in the pre-disposal stage, the blood slides can still be accurately analyzed with microscopy, and this implies that the quality of the captured data was still high.

4.2. Sample preparation

Before the blood slides were selected for analysis, an inspection was done to ensure that only slides with good physical quality were considered. Some defects to physical quality include cracks in the slide glass and foreign artefacts that are not expected to be on the slides. Only slides that passed the quality checks were considered for data capture.

The preparation of a blood slide for data capture mainly involves staining with emulsion oil to improve visibility under the microscope. Once stained, the thick blood slides are ready for data capture. For the thin blood slides, an extra step involves spreading the blood on the slide, after which it can then be used to capture the data.

4.3. Data collection tools

The image data was collected using smartphones attached to an Olympus CX-23 Microscope² which was used to view the blood slides. To use the smartphone with the microscope, a 3D-printable adapter was designed to house the phone and attach it to the eyepiece of the microscope as shown in Fig. 1. The phone adapter ensures the stability of the smartphone when capturing images and standardizes the use of different phone types and designs for the same task of capturing images on the microscope. The adapter's design also ensures that data can be captured using different microscopes with minimal adjustment to the process (Tables 2 and 3).

Three different smartphone models—the Samsung S8+, Redmi 7, and Redmi 8—were used to capture the images in this dataset. The selection of the smartphones was based on two main considerations: the variation in the dataset, which is important for developing generalizable models with the images, and the fact that these models are some of the most readily available and accessible on the market. (Figs. 2 and 3).

Image capture on the smartphones was done using a publicly available Android application on the Google Play Store called MegaZoom.³ This application was selected to standardize the environment under which the images were captured on different phones since different models will have varying camera characteristics like lens resolution and software-based enhancements.

² [https://www.olympus-lifescience.com/en/microscopes/upright/cx23/#!cms\[focus\]=cmsContent1392](https://www.olympus-lifescience.com/en/microscopes/upright/cx23/#!cms[focus]=cmsContent1392).

³ <https://play.google.com/store/apps/details?id=com.just4funtools.megazoomcamera.xzoom&hl=en&gl=US>.



Fig. 1. A microscope with a phone mounted using the 3D printed adapter on the left and a 3D printer to its right.

Table 2
Number of images captured for thick and thin slide types by phone type.

Slide Type	Device	Number of Images
Thick	Samsung s8	1368
	Redmi 8	1086
	Redmi 9	546
Thin	Samsung s8	500
	Redmi 8	500

Table 3
Number of objects annotated per class in the thick and thin blood slide images.

Country	Category	Images	Classes
Uganda	Thick	3000	Trophozoite- 20392
			WBC - 4127
	Thin	1000	Parasitized cell- 1093
			Trophozoite - 53
			WBC - 984
			Artefacts - 7

4.4. Data collection

The data collection involved three main processes: microscope setup, phone setup and data validation. The Olympus CX23 Microscope was set up as is usually done by the lab technicians for analysis of blood slides to check for the presence of malaria parasites. This set-up included a magnification of 1000×, which is a product of the eyepiece lens and nosepiece lens under which the blood slide is directly placed. This set-up applied to both the thick and thin blood slides. Additionally, the lab technician turned on the illumination under the slide, which is usu-

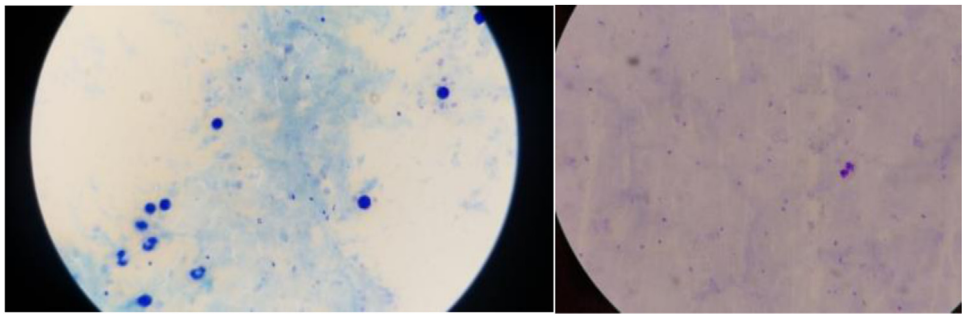


Fig. 2. Sample thick blood slide images captured with Redmi 8 phone (left) and Samsung s8+ phone (right).



Fig. 3. A screenshot showing the VGG image annotator tool.

ally varied from levels four to six on the specific microscope. Still, it should be noted that this can vary based on individual expert preference. The lab technician then performed a visual inspection through the eyepiece to ensure that they can see the different objects of interest on the blood slide and confirming that the microscope has been correctly set up. The setup for the microscope is usually performed once for a data collection session with different blood slides, and adjustments will be made at the expert's discretion.

In the next stage, the 3D-printed plastic adapter was attached to the microscope via the eyepiece, and the phone is placed on the adapter. The lab technician then launched the MegaZoom application to launch the camera for taking pictures. The adapter provides adjustments for positioning the phone's camera lens directly over the eyepiece to ensure that the lab technician has a clear view of the blood slide on the phone screen. Tests were conducted using the three smartphones to identify the ideal settings to use within the MegaZoom application, and three lab technicians who conducted the test agreed that using a 14 % zoom level in the MegaZoom application produced very clear images. The rationale for the selection of the zoom level was to have images from which the objects of interest could be easily identified for data annotation.

For the image capture, the experts agreed that for each slide, one could capture up to 50 images by adjusting different fields of view of the blood slide under the microscope.

4.5. Data annotation

Experts labelled the collected data to show the objects of interest in the images. The goal of the labelling was to create datasets that could be used to conduct machine learning experiments to detect different specimens in the blood slide images. Therefore, the labelling involved drawing bounding box annotations on the objects of interest in the images.

The tool of choice for data annotation was the VGG Image Annotator (VIA) [9], an open-source annotation tool used for labelling images and video. The VIA tool supports annotation of different types, including bounding boxes, which was the choice for annotation of the collected images. This tool also supports the annotation of cloud-hosted files, which was a key requirement as the annotators of the data were not physically stationed in one place and eliminated the challenge of manually moving data files between devices that were going to be used by the annotators.

Before the data annotation, the collected data was validated to check for common issues like duplication. A visual inspection was done to ensure that the images represented blood slides and to get rid of blurry images. The validated data was then hosted on a cloud bucket, and access URLs were generated for each file. Annotation projects were created in the VIA tool, and these projects defined the labels that will be used for annotation and the images to be annotated. Each annotation project had 50 images to be annotated, as this was deemed manageable by the annotators.

The next step involved selecting experts to annotate the images and consequent training. Two microscopists who regularly analyse blood slide images at Kiruddu National Referral Hospital were selected to do the annotation due to their expertise in microscopy and, therefore, are skilled in identifying different objects of interest in the blood. The other reason for selecting 2 experts was to conduct validation by getting the Inter Annotator Agreement for a subset of the images. The two experts were trained on how to use the VIA tool and feedback from the experts was taken to improve the setup of the annotation projects. The main improvement that the experts requested was to show differences in the annotations of different objects of interest, which was done as shown in Fig. 4.

For the annotation of the thick blood slide images, the objects of interest were malaria trophozoites and white blood cells. These were selected because they are the main objects ob-

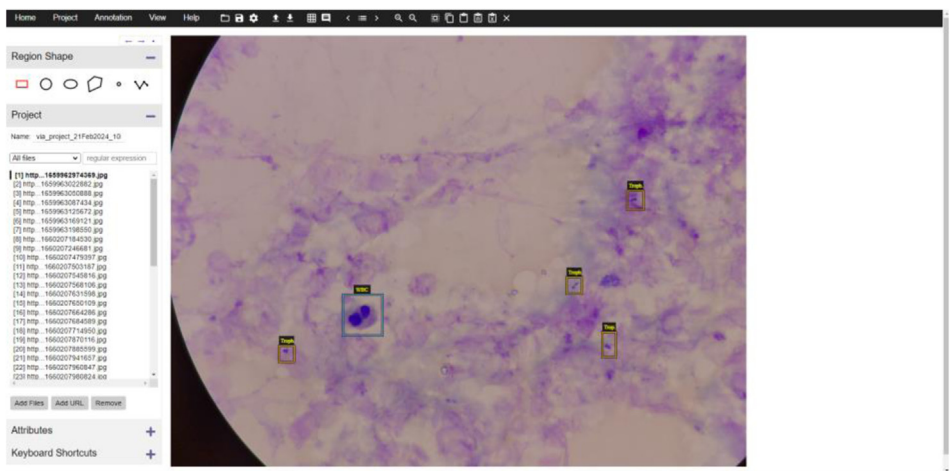


Fig. 4. Annotation with the VGG Image Annotator showing four annotated objects (Three trophozoites with yellow bounding boxes and a white blood cell with the blue bounding box) in a thick blood slide image. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

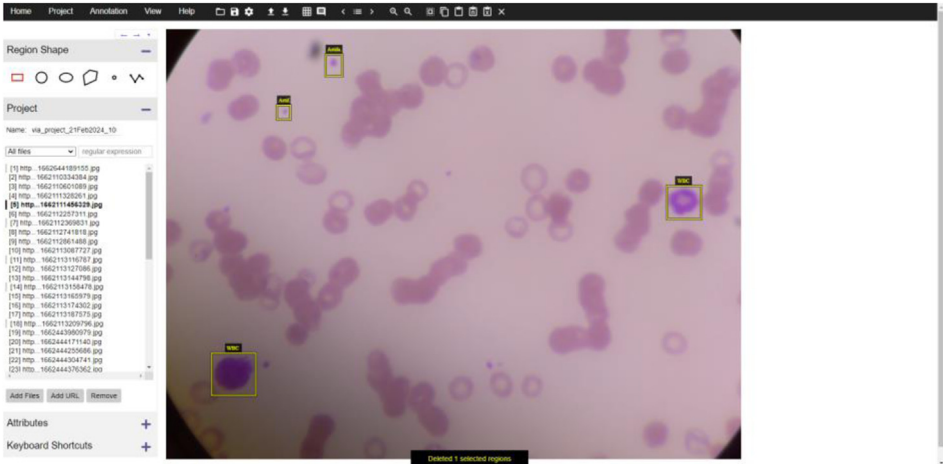


Fig. 5. Annotation with the VGG Image Annotator showing four annotated objects (two white blood cells and two artefacts) in a thin blood slide image.

served by microscopists during the determination of the presence of malaria and for the quantification of parasitaemia.

For the thin blood slide images, the objects of interest were the malaria parasite stage (Gametocyte or Trophozoite), white blood cells, parasitized cells (cells with a malaria trophozoite) and artefacts (Fig. 5).

Limitations

The main limitation of this dataset is in relation to the thin blood slide images regarding variation in the species of malaria and the life-cycle stages of the malaria parasites. Over 95 % of the images captured showed only the Trophozoite stage for the parasite life cycle and only one the Falciparum species of the parasite.

Ethics Statement

We have read the ethical requirements for publication in Data in Brief and confirm that the current work does not involve human subjects, animal experiments, or any data collected from social media platforms.

Credit Author Statement

Rose-Nakasi: Conceptualization, Methodology, Investigation, Supervision, Writing (Original Draft, Editing and Review), Resources, Funding acquisition, Validation, Project Administration. **Joyce Nakatumba Nabende:** Conceptualization, supervision, Methodology, Writing (Editing and Review). **Francis Jeremy- Tusubira:** Methodology, Writing (Original draft, Editing and Review), Software, Data Curation, Resources. Aloyzius Lubowa **Bamundaga:** Investigation, Data Curation. **Alfred-Andama:** - Conceptualization, Methodology, Investigation, Data Curation.

Data Availability

A Dataset of Blood Slide Images for AI-based Diagnosis of Malaria (Original data) (Dataverse).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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